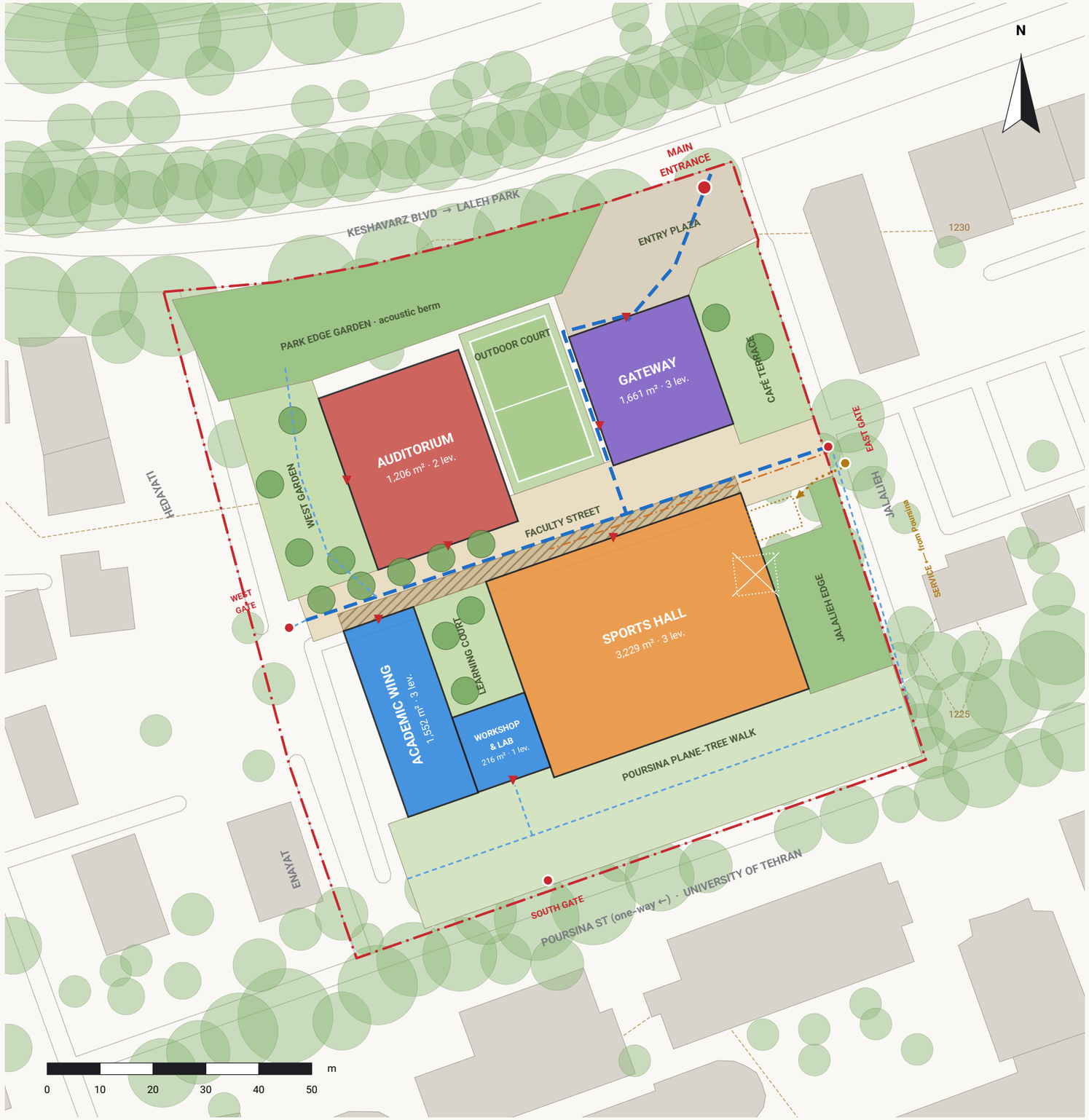
01

CONCEPT SITE PLAN — "FACULTY STREET"

Proposal drawn directly over the supplied DXF survey · boundary = road edges of the block marked by the user · 1:500 at A1 (scaled here)

سایت‌پلان ایده — خیابانِ دانشکده



CONCEPT

The old Zare' service lane is turned into a shaded public street that crosses the block from Jalalieh to the western lanes. The faculty is organised as rooms along this street: public and event uses face Laleh Park, the sports hall faces the University, learning faces the quiet residential west.

کوچه‌ی سرویس زارع به یک خیابان عمومی سایه‌دار تبدیل می‌شود که بلوک را از جلالیه تا کوچه‌های غربی قطع می‌کند. دانشکده مثل اتاق‌هایی در دو سوی این خیابان سازمان می‌یابد: کاربری‌های عمومی و رویدادی رو به پارک لاله، سالن ورزشی رو به دانشگاه، و آموزش رو به لبه‌ی آرام مسکونی غرب.

LEGEND

- Sports hall (B1)
- Auditorium (B4)
- Academic wing + workshop / lab (B3, B2)
- Gateway: café · welfare · assembly hall · admin · presidency (B5)
- Plaza / Faculty Street
- Courts and gardens
- Green buffer — existing plane trees kept
- Site boundary (user-marked, on the road edges)
- Primary pedestrian route
- Secondary pedestrian route
- Emergency access (no public traffic)
- Service route + loading bay
- Site entrance
- Building entrance
- Proposed tree (pale circles = surveyed trees, DXF)
- Existing structure removed (56 m²)
- 5 m contours from the DXF (1225 / 1230)

KEY NUMBERS

Site (user boundary)	**13,845 m²**
Program net / gross	**5,759 / 7,864 m²**
Ground-floor footprint	**4,313 m² (31 %)**
Gross / site area	**0.57**
Open space	**9,532 m² (69 %)**
Height	**1–3 levels (hall ≈12 m, sunk ≈1.5 m)**

02

CONCEPT INFERENCE

استنتاج ایده از تحلیل سایت

Site findings → design opportunities → design principles → architectural concept → spatial organisation

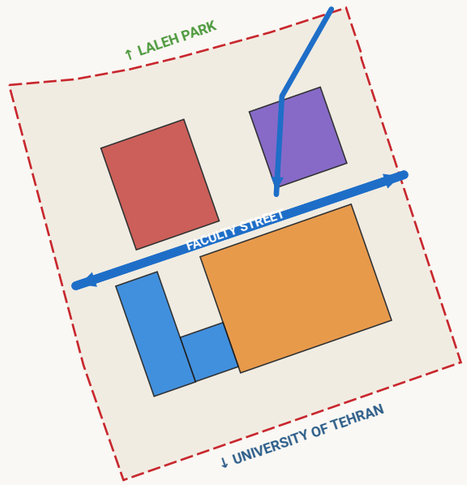
SITE FINDINGS یافته‌ها	OPPORTUNITIES / CONSTRAINTS فرصت / محدودیت
F1 Site sits between Laleh Park (35 ha, north) and the University of Tehran central campus (≈21 ha, south) Atlas 01–03 · OSM · documented	OPPORTUNITY A threshold building: public face to the park, institutional face to the campus
F2 Jalalieh (dead end for cars) + Zare' lane are the only pedestrian link park ↔ campus; metro Boostan-e Laleh L6 ≈ 280 m Atlas 05, 06, 21 · OSM · documented	OPPORTUNITY Keep the link public: Zare' becomes the Faculty Street; main entry at Keshavarz × Jalalieh
F3 No side opens onto an arterial; cars reach the block only from one-way Poursina via Jalalieh / Enayat Atlas 05–06 · documented	CONSTRAINT No traffic through the site; service and emergency via Jalalieh from Poursina
F4 Keshavarz Blvd: divided boulevard, buses, fire-station sirens; wind and dust channelled along it (no dB measured) Atlas 14–15 · documented	CONSTRAINT North green / acoustic buffer; the blank wall of the auditorium faces north
F5 West edge: fine-grain residential with blank party walls (sensitive neighbour) Atlas 07–09, 19 · documented	CONSTRAINT Quiet, lower west edge: learning + garden; no outdoor sport or night lighting toward it
F6 South and east: institutional fabric (UT, TUMS, Institute of Archaeology, History of Science) Atlas 07, 18 · documented	OPPORTUNITY Large-span sports hall toward the south-east institutional edge
F7 Mature plane-tree rows on Keshavarz, Poursina and Jalalieh (55 surveyed trees in or within 4 m of the boundary) Atlas 10 · DXF layer TPX_VEGETATION_TREES_INDIVIDUAL · documented	OPPORTUNITY Keep every edge row; no footprint inside a tree crown
F8 Ground falls north → south: ≈3 m (SRTM, atlas 11) but ≈5 m between the 1230 and 1225 contours of the DXF Atlas 11 · DXF TPX_CONTOUR_LINES_5M · hypothesis — verify	OPPORTUNITY Put the tallest volume (hall) at the low south edge, floor ≈1.5 m below the street; survey needed
F9 Semi-arid cold climate (BSk): hot sunny summers, strong west sun, winter inversions; sun path computed for the site Atlas 12–13 · documented	OPPORTUNITY + CONSTRAINT Courts open south; closed, shaded west facades; outdoor court with a N–S long axis
F10 Winds: SW (annual) and W (brief), plain↔mountain daily flows; summer dust; calm winter inversions Atlas 14 · documented	OPPORTUNITY + CONSTRAINT Buildings shelter the courts from the W; the Faculty Street runs E–W for cross ventilation of the street
F11 Views: north to the park (upper floors), south to the campus and Poursina plane trees; blank walls W and E Atlas 09, 16 · documented	OPPORTUNITY Admin / presidency on the top floor of the corner building, looking to the park
F12 User boundary = 13,846 m² (road edges of the block) vs program 7,864 m² gross DXF · brief · calculated	OPPORTUNITY Fits on 1–3 levels with ≈31 % ground coverage; no FAR / coverage limit known
F13 The existing surface parking is lost; local kerbside parking is already under pressure Atlas 19 · documented	CONSTRAINT Rely on metro + campus; drop-off and accessible bays on Jalalieh (to confirm)
F14 Brief item "سایت" (89 / 66 m²) has no stated function Brief · hypothesis — verify	CONSTRAINT Keep it as an unresolved, reserved space; do not reinterpret

DESIGN DRIVERS محرک‌های طراحی	PRINCIPLES اصول
THRESHOLD Park ↔ University: the faculty is a public passage, not an enclosure	1 **Open the block: one public street from Jalalieh to the western lanes.**
FACULTY STREET Zare' lane, the only cross-block route, becomes the spine every building opens onto	2 **Public faces the park, sport faces the university, learning faces the quiet west.**
FOUR EDGES N public & noisy · S institutional · W quiet & private · E pedestrian	3 **Buffer, do not wall: keep the plane trees, add a planted berm on Keshavarz.**
KEEP THE TREES The plane-tree rows define the edges; nothing is built inside a crown	4 **Every building enters from the shaded Faculty Street.**
BULK WHERE IT FITS Big, tall volumes low and south-east; the 3-level landmark at the park corner	5 **Large volumes low and south; small volumes stack at the corner.**

Drivers combine F1–F14; see the O/C chips (green = opportunity, red = constraint).

CONCEPT → ORGANISATION

ایده ← سازمان فضایی



FACULTY STREET خیابانِ دانشکده

- North band (park side): auditorium · outdoor court · gateway (café, welfare, assembly hall, admin, presidency)
- Faculty Street: shared, shaded spine; every entrance opens onto it
- South-east (campus side): sports hall + workshop / lab, service at the street's east end
- South-west (quiet): academic wing + library around the Learning Court
- Edges: plane-tree rows kept; acoustic berm on Keshavarz; west garden as buffer

THREE CONCEPT ALTERNATIVES

سه گزینه‌ی ایده

Each tested on the real boundary · scores 1–5 against the seven studio criteria · selected: A



A · FACULTY STREET — خیابان دانشکده

Idea — Zare' lane becomes a public E–W street; buildings are rooms along it, grouped by edge character.

Evidence — F1 F2 F4 F5 F6 F7

Placement — Auditorium + gateway on the park side; hall in the SE toward the campus; academic wing on the quiet west.

Organisation — Public/events N · sports SE · learning SW · admin at the main corner.

Circulation — One spine (E–W) + park-corner plaza; perimeter walks under the existing trees; service only at the E end.

Built / open — Sequence of plaza → street → courts; edges stay green.

+ — Keeps the only park–campus pedestrian link public; every building has a clear address; phasing by building.

- — Long spine needs shade and night management; hall bulk visible from the street.



B · COURTYARD FACULTY — دانشکده‌ی حیاط‌مرکزی

Idea — A perimeter ring of buildings around one large central sports-and-learning courtyard.

Evidence — F4 F9 F10

Placement — Bars on all four edges; hall on the south side; court in the middle.

Organisation — Program wraps the court; entrances from the corners.

Circulation — Internal ring corridor around the court; one controlled gate.

Built / open — One protected court; edges become building fronts.

+ — Strong identity; sheltered microclimate; easy control.

- — Closes the park–campus link and the Zare' lane; builds against the west residents; edge trees under pressure; poor expansion.



C · LANDSCAPE PLINTH — سکوی منظر

Idea — Hall and auditorium sunk under a public landscape deck that continues Laleh Park toward the campus; a thin bar sits on top.

Evidence — F1 F7 F8

Placement — Large volumes below the deck; academic + admin bar along the west and north on the deck.

Organisation — Below: sports + events; above: learning + admin.

Circulation — Deck is public; ramps and stairs at the corners; service ramp from Jalalieh.

Built / open — Maximum public green; the roof is the park.

+ — Least visible bulk; maximum open space; strong landscape identity.

- — Deep excavation and long spans under soil load; daylight and fire egress for the hall; high cost and risk at concept stage.

EVALUATION

	Response to site	Functional clarity	Spatial relationships	Circulation efficiency	Open-space quality	Future expansion	Architectural identity	TOTAL / 35
A · Faculty Street	■■■■■	■■■■■	■■■■□	■■■■■	■■■■□	■■■■□	■■■■□	31
B · Courtyard Faculty	■■■□□	■■■■□	■■■■□	■■■□□	■■■□□	■■□□□	■■■■□	23
C · Landscape Plinth	■■■■□	■■■□□	■■■■□	■■■□□	■■■■■	■■■□□	■■■■■	27

Why A: it is the only option that keeps the park–campus pedestrian link public (F1, F2), puts each use on the edge whose character fits it (F4–F6) and keeps every existing tree row (F7) at a low technical risk. B closes the link and builds against the west residents; C scores highest on landscape but depends on deep excavation and long buried spans that cannot be justified at this stage.

04

FUNCTIONAL RELATIONSHIPS

Hypotheses tested on the plan: the matrix, the bubble diagram and the site plan describe the same arrangement

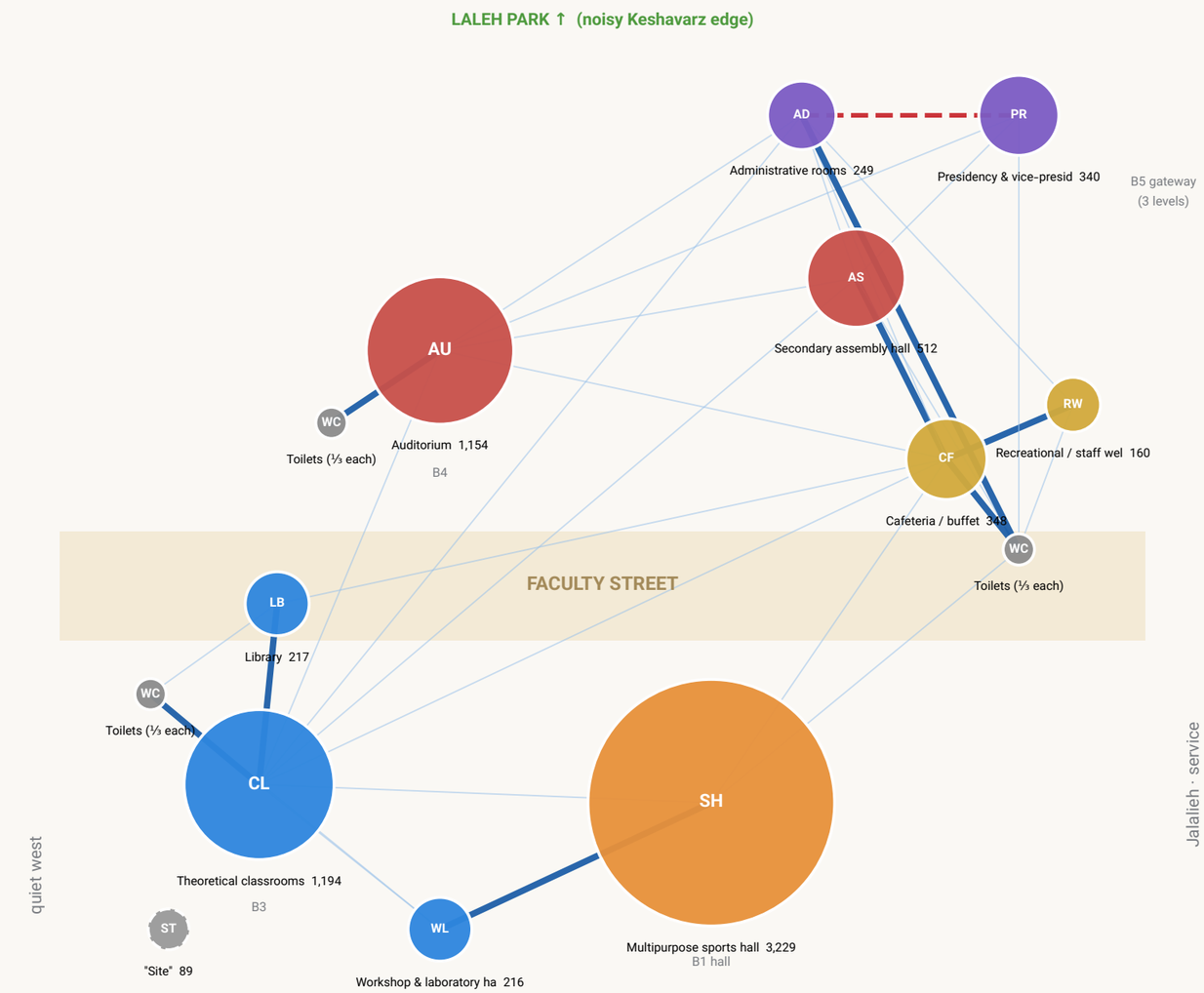
ماتریس و دیاگرام ارتباطات

RELATIONSHIP MATRIX (gross m² at left)

		SH	WL	CL	LB	ST	AU	AS	CF	RW	AD	PR	WC
SH	Multipurpose sports hall	3,229		S	M	W	?	W	W	M	W	W	W	M
WL	Workshop & laboratory hall	216			M	W	?	W	W	W	W	W	W	M
CL	Theoretical classrooms	1,194				S	?	M	M	M	W	M	W	S
LB	Library	217					?	W	W	M	W	W	W	M
ST	"Site"	89						?	?	?	?	?	?	?
AU	Auditorium	1,154							M	M	W	M	M	S
AS	Secondary assembly hall	512								S	W	M	M	M
CF	Cafeteria / buffet	348									S	M	W	S
RW	Recreational / staff welfare	160										M	W	M
AD	Administrative rooms	249											R	S
PR	Presidency & vice-presidency	340												M
WC	Toilets	156												

S	Strong — directly adjacent
M	Moderate — conveniently connected
W	Weak — separated where appropriate
R	Restricted — controlled access
?	Unresolved — "سایت" function unknown

BUBBLE DIAGRAM (area ∝ gross · lines = S / M / R)



05 ZONING · CIRCULATION

پهنه‌بندی و گردش

Access levels and noise (left) · movement system, gates and service (right) — both on the real boundary





06

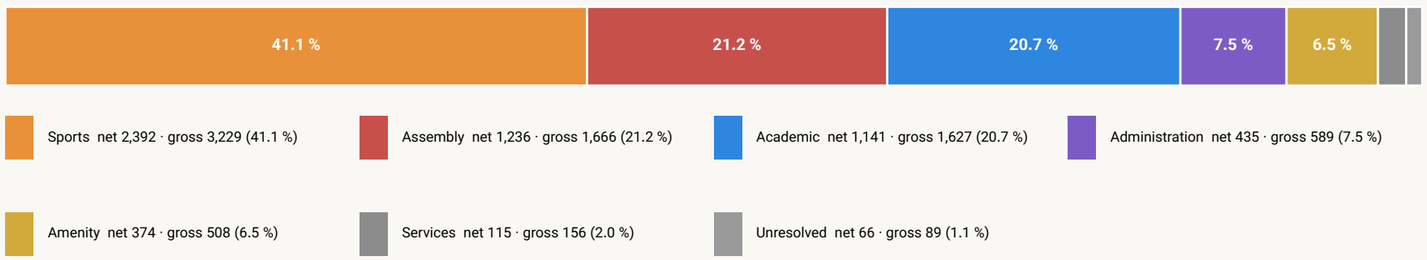
PROGRAM · STACKING · RATIONALE

جدول مساحت‌ها و منطق طرح

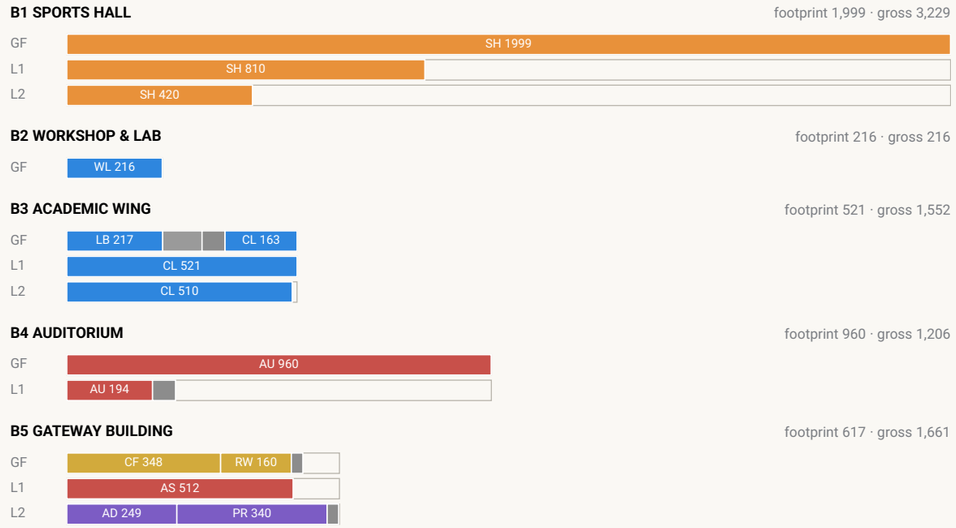
Brief areas unchanged · gross = planning target · ground-floor footprint ≠ gross floor area (levels shown)

Functional space	فضا	Net	Gross	Group	Placed in (building · level · m²)
Theoretical classrooms	کلاس‌های تئوری	820	**1,194**	Academic	B3 GF (163) · B3 L1 (521) · B3 L2 (510)
Multipurpose sports hall	سالن ورزشی چندمنظوره	2,392	**3,229**	Sports	B1 GF (1999) · B1 L1 (810) · B1 L2 (420)
Auditorium	آمفی‌تئاتر	855	**1,154**	Assembly	B4 GF (960) · B4 L1 (194)
Library	کتابخانه	161	**217**	Academic	B3 GF (217)
Secondary assembly hall	سالن اجتماعات	381	**512**	Assembly	B5 L1 (512)
Workshop & laboratory hall	سالن کارگاه و آزمایشگاه	160	**216**	Academic	B2 GF (216)
"Site" (سایت) — function unresolved	سایت (نامشخص)	66	**89**	Unresolved	B3 GF (89)
Administrative rooms	اتاق‌های اداری	184	**249**	Administration	B5 L2 (249)
Presidency & vice-presidency	ریاست و معاونت‌ها	251	**340**	Administration	B5 L2 (340)
Recreational / staff welfare	تفریحی / رفاهی	116	**160**	Amenity	B5 GF (160)
Cafeteria / buffet	بوفه	258	**348**	Amenity	B5 GF (348)
Toilets	سرویس‌های بهداشتی	115	**156**	Services	B3 GF (52) · B4 L1 (52) · B5 GF (26) · B5 L2 (26)
TOTAL (12 items)		**5,759**	**7,864**		gross / net = 1.366

FUNCTIONAL GROUPS — share of the total gross area



STACKING — floors drawn to scale of area (bar = footprint)



DESIGN RATIONALE

The block is the only pedestrian passage between Laleh Park and the campus (F1, F2). Turning the Zare' lane into the Faculty Street keeps that passage public and gives every building one shared address.

Each use goes to the edge whose character fits it: events and the gateway to the busy park side (F4), the long-span hall to the institutional south-east (F6), learning to the quiet west (F5).

The program needs only ≈31 % ground coverage on 1–3 levels, so all three plane-tree edges are kept (F7) and the courts can be oriented for winter sun and summer shade (F9–F10).

The hall sits at the low south edge, sunk ≈1.5 m (F8), so its ≈12 m volume reads lower than the 3–8-storey context.

ASSUMPTIONS · UNRESOLVED

- Brief item "Site" (66 / 89 m²): function unknown — reserved in the academic wing GF, not reinterpreted.
- Boundary from the road edges marked by the user; lane ownership (Hedayati, Enayat) to verify.
- Levels: DXF contours (≈5 m fall) vs SRTM (≈3 m) — topographic survey needed.
- No FAR / coverage / height limit supplied; none assumed. No on-site parking provided.